

CHARAN ROMI

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PROFILE

Full-stack developer specializing in React/Next.js and Node.js, with production experience building REST APIs, PostgreSQL data models, and role-based systems in a healthcare setting. A final-year software engineering student available for full-time work now, balancing final coursework alongside employment.

SKILLS

Languages: JavaScript, TypeScript

Frontend: React.js, Next.js, Tailwind CSS, Zustand, Context API

Backend: Node.js, Express.js, RESTful API, JWT authentication & authorization

Databases: PostgreSQL, MySQL, MongoDB, Prisma, TypeORM

Tools & DevOps: Git, Docker, GitHub Actions, AWS S3, Postman

Practices: Agile/Scrum, OOP, unit testing, Jest, SDLC, CI/CD

WORK EXPERIENCE

Full-Stack Developer Intern - Vectorium Labs

Jul 2025 – Mar 2026

- Developed and shipped more than 15 production REST API endpoints (Express.js, PostgreSQL) with input validation and role-based access control for users, facilities, resources, reports, and payment modules in a healthcare management system.
- Integrated the PointClickCare healthcare API and webhooks to automate data synchronization and eliminate manual re-entry of patient and payment records between systems.
- Designed and shipped a responsive Next.js admin dashboard consolidating facilities, resources, users, and reports into a single interface, used by the administrative team.
- Collaborated in daily stand-ups and sprint planning within a five-member Agile team; maintained Git workflows across concurrent feature branches and resolved merge conflicts before merging into the production branch.

EDUCATION

BSc (Hons) in Software Engineering - Open University of Sri Lanka

2022 – Present

Final-year student. Final examinations: January 2027.

PERSONAL PROJECTS

Hyper Play - Multi-Vendor Sports E-Commerce Platform (2025) | [GitHub](#) | [Live](#)

React.js, Node.js, Express.js, PostgreSQL, Docker, Redis, Stripe, AWS S3

- Developed separate customer, manufacturer, and administrator interfaces with role-based access and functionality.
- Designed an 8-state order lifecycle with enforced role-based transition rules, eliminating invalid or unauthorized order-status changes.
- Implemented an inventory reservation system and payment-confirmed stock updates to prevent overselling.
- Built an automated manufacturer assignment workflow that selects an active manufacturer with available capacity and assigns paid orders transactionally.
- Integrated Stripe payments (test mode), AWS S3 media storage, Brevo notifications, and Redis caching.
- Containerized the application with Docker and implemented a CI/CD workflow using GitHub Actions and Render.

Edu Nova — Learning Management System (2026) | [GitHub](#) | [Live](#)

Next.js, TypeScript, Node.js, Express.js, MongoDB, Docker

- Built an Adaptive LMS with dedicated student, instructor, and administrator interfaces, each scoped to its own permission set via JWT-based auth and role-based authorization.
- Designed REST APIs and MongoDB data models spanning institutes, users, courses, and enrollments.
- Developed an AI-driven curriculum generator using the Groq API and deterministic prerequisite graph sorting to transform student career goals into structured, ordered learning paths.
- Integrated the Groq API to parse student requirements in natural language and recommend relevant courses.

Nutrio AI — Personalized Nutrition & Meal Planning Platform (2026) | [GitHub](#)

React Native, TypeScript, NestJS, PostgreSQL

- Developed an AI nutrition platform that generates personalized, culturally tailored meal plans for Sri Lankan users based on biometrics, dietary restrictions, and budget.
- Engineered an AI meal generation pipeline integrating the Google Gemini API with structured JSON schema outputs, translating user profile data (BMR/TDEE, allergies, cuisine preferences, LKR budget) into multi-day meal plans persisted across a normalized meal-plan/meal-item schema.
- Built grocery list automation and an adaptive feedback system (ratings, reason tags) to support future personalization.